A Fixed-Point BSDE Framework to Solve Household Saving and Multi-Country Asset Pricing Models

Abstract

This technical note develops a block-wise Backward Stochastic Differential Equation (BSDE) evaluation framework for continuous-time stochastic general equilibrium models. The core principle is that BSDEs can only reliably approximate value functions or asset price functions when all policy rules and state drift dynamics are held fixed; policy or equilibrium rate updates are treated as outer block iterations separated from inner BSDE training. We first present a canonical two-state stochastic neoclassical growth toy model with Ornstein–Uhlenbeck TFP shocks, solved via BSDE-based Howard policy iteration. We then extend the block BSDE method to a three-country heterogeneous-agent international macro model with expert capital owners and household savers, where we alternate BSDE asset price evaluation and global risk-free rate updating via block Gauss–Seidel iteration. We provide full model dynamics, BSDE residual loss formulations, iterative algorithms, closed-form equilibrium restrictions, and analytical comparative static results for structural parameters. Benchmark comparisons against standard HJB finite difference solvers and cross-parameter sensitivity experiments are discussed.

1 Toy Continuous-Time Stochastic Consumption-Saving Model

This section introduces a minimal representative-agent stochastic growth environment to illustrate the core BSDE policy iteration paradigm: freeze the consumption policy, train BSDE critic networks for value V and martingale diffusion Z, then update consumption via Hamiltonian first-order conditions (FOC).

The policy iteration framework based on backward measurability loss (BML) is first proposed in Wang and Ni 2022 for decoupled linear FBSDEs in stochastic growth models.

For fully coupled nonlinear FBSDE systems, Wang, Ni, and Li 2023 generalize the BML loss function and provide rigorous fixed-point convergence theory.

1.1 Model Primitives and State Dynamics

The model features two Markov state variables: private capital stock k and stationary aggregate productivity shock z. Instantaneous output is $y = e^z k^\alpha$. Capital accumulation follows

$$dk = [e^z k^\alpha - \delta k - c(k,z)]\, dt,$$

where $c(k,z)$ denotes instantaneous consumption flow. Productivity evolves as a mean-reverting Ornstein–Uhlenbeck (OU) process driven by a standard Brownian motion W_t:

$$dz = \kappa_z\,(\bar{z} - z)\, dt + \sigma_z dW_t.$$

The household maximizes infinite-horizon discounted CRRA utility:

$$V(k,z) = \sup_{\{c_t\}} \mathbb{E}\left[\int_0^\infty e^{-\rho t} u(c_t) dt \,\middle|\, k_0 = k,\ z_0 = z \right],$$

with utility kernel

$$u(c) = \frac{c^{1-\gamma} - 1}{1-\gamma}, \quad \gamma > 0, \gamma \neq 1.$$

1.2 Baseline Calibration

Table 1: Toy Model Structural Parameters

Parameter	Description	Value
α	Capital income share	0.36
δ	Capital depreciation rate	0.08
γ	Coefficient of relative risk aversion	2.00
ρ	Subjective time discount rate	0.05
κ_z	TFP mean reversion speed	0.30
$\bar{z}$	Long-run mean TFP shock	0
σ_z	TFP diffusion volatility	0.023
Δt	Discrete Euler BSDE time step	0.005

Numerical state grid bounds for training and interpolation:

$$k \in [0.05,\ 30], \qquad z \in [-0.12,\ 0.12].$$

1.3 Fixed-Policy BSDE Formulation

Conditional on a frozen consumption policy $c(k,z)$, the value process V_t solves the backward SDE

$$dV_t = -\left[u(c_t) - \rho V_t\right] dt + Z_t dW_t,$$

where Z_t is the BSDE martingale diffusion term capturing exposure to Brownian productivity shocks. Discretize forward state transition one step to future state $S_{t+\Delta t} = (k', z')$. Define the one-step BSDE residual

$$R_t = V_t + \left[\rho V_t - u(c_t)\right] \Delta t + Z_t \Delta W_t - V(S_{t+\Delta t}),$$

where $\Delta W_t \sim \mathcal{N}(0, \Delta t)$ is the discrete Brownian increment. The network training loss minimizes the expected squared residual:

$$\mathcal{L}_{\text{BSDE}} = \mathbb{E}\left[R_t^2\right].$$

Crucially, consumption c is held fixed throughout inner BSDE loss minimization; the network only approximates V and Z given the pre-specified policy.

1.4 BSDE Policy Iteration Algorithm (Howard Iteration)

The algorithm separates policy evaluation and policy improvement into disjoint outer rounds, with no joint optimization over consumption, value, and diffusion within a single training loop:

(1) Initialize a feasible starting consumption guess: $c^0(k,z) = 0.1 \cdot e^z k^\alpha$.

(2) Policy Evaluation (Inner BSDE Step): Freeze policy c^n. Train a joint feedforward critic network mapping $(k,z) \mapsto (V^n, Z^n)$ by minimizing $\mathcal{L}_{\text{BSDE}}$ over batches of simulated state trajectories.

(3) Policy Improvement (Outer Update Step): Compute the marginal value of capital $V_k^n = \partial V^n / \partial k$. Apply the Hamiltonian FOC $u'(c) = V_k$ to solve for updated consumption:

$$c^{n+1} = (V_k^n)^{-1/\gamma}.$$

Numerical safeguards (damped updates, consumption lower bounds, feasibility clipping to ensure positive investment) are applied to avoid policy explosion near grid boundaries. The updated policy c^{n+1} is stored as a grid-interpolated object for the next iteration.

(4) Iterate for fixed total rounds (baseline experiment: 20 outer iterations, 20,000 BSDE training steps per evaluation).

The iterative mapping is compactly written as

$$c^n \mapsto \left(V^{c^n}, Z^{c^n}\right) \mapsto c^{n+1}.$$

At the theoretical fixed point $c^* = \mathcal{G}(V^{c^*})$, the policy is self-consistent with its associated value function. However, neural approximation error, inaccurate state derivatives near grid edges, and non-contractive outer update maps break standard theoretical convergence guarantees for exact Howard iteration; the continuous-time HJB finite difference solver serves as a benchmark validation tool.

1.5 Numerical Output Pipeline

After each outer policy iteration round, the implementation exports:

(i) Slices at steady-state productivity $z = 0$: plots of $V(k, 0)$, $c(k, 0)$, marginal value $V_k(k, 0)$, BSDE diffusion $Z(k, 0)$ (saved as `toy_bsde_round_XX.png`).

(ii) Convergence charts tracking cross-grid maximum absolute deviations of V and c across all 20 rounds.

(iii) Benchmark overlay figure `toy_hjb_bsde_overlay.png` comparing BSDE policy iteration against standard HJB finite difference Howard iteration.

(iv) Full grid numerical dataset serialized as compressed `.npz` files for post-simulation quantitative analysis.

1.6 Results

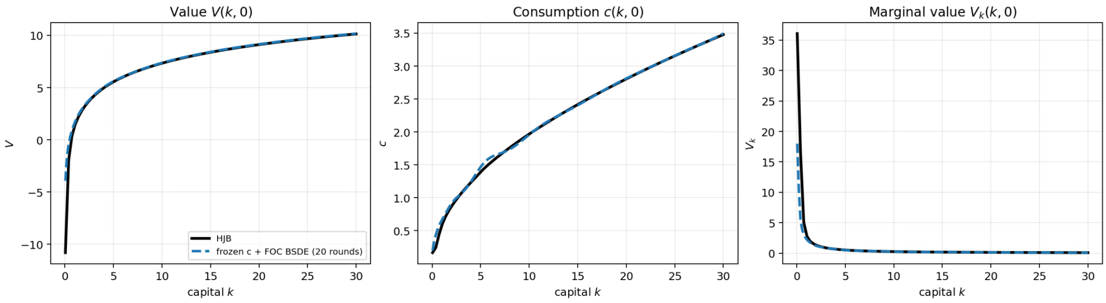


Figure 1: Comparison between exact HJB solution (solid black line) and frozen consumption BSDE-FOC with 20 iterations (dashed blue line). Left panel: value function $V(k, 0)$; Middle panel: consumption policy $c(k, 0)$; Right panel: marginal value $V_k(k, 0)$. Horizontal axis: capital k.

2 Three-Country Macro Finance Model in Huang 2022

This paper extends the fixed-drift BSDE evaluation logic to a multi-country framework with two agent types: capital-owning experts and wealth-saving households. The core block iteration structure is analogous to the toy model—fix a subset of equilibrium objects, run BSDE evaluation, then update the remaining equilibrium price block—though the update step here targets a global risk-free rate rather than a household consumption policy.

2.1 Economic Environment

Three symmetric countries indexed $i \in \{1, 2, 3\}$ with segmented physical capital markets:

- Expert agents hold all domestic productive capital; cross-border direct capital investment is prohibited.
- Households trade a single common international risk-free bond, equalizing the global equilibrium interest rate $r(S)$ across all regions.
- Household preferences are logarithmic, implying aggregate consumption is proportional to total wealth with constant consumption rate ρ.

The aggregate state vector S collects wealth distribution shares:

$$S = (\eta_1, \eta_2, \eta_3, \zeta_1, \zeta_2), \qquad \zeta_3 = 1 - \zeta_1 - \zeta_2,$$

where η_i = expert wealth share in country i, ζ_i = country i's share of total world wealth.

2.2 Capital, Investment, and Asset Price Dynamics

Let q_i denote the market price of installed physical capital in country i. Investment adjustment costs $\Phi(\iota) = \frac{1}{\psi} \ln(\psi\iota + 1)$ produce investment flow

$$\iota_i = \frac{q_i - 1}{\psi}.$$

Country-i capital stock evolves with constant fundamental volatility σ:

$$\frac{dK_i}{K_i} = (a - \iota_i)\, dt + \sigma dW_i.$$

The capital price follows a diffusion process with drift $\mu_{q,i}$ and volatility matrix σ_q (row $i = \sigma_{q,i\cdot}$):

$$\frac{dq_i}{q_i} = \mu_{q,i} dt + \sigma_{q,i\cdot} d\boldsymbol{W}.$$

Define total capital return exposure for country i:

$$\sigma^K_{i\cdot} = \sigma_{q,i\cdot} + \sigma \boldsymbol{e}_i^\top,$$

where $\boldsymbol{e}_i$ is the unit vector selecting the i-th Brownian shock.

2.3 Expert Euler Equation and Price Drift Restriction

With global interest rate $r(S)$, the expert's asset pricing Euler condition delivers a expression for the price drift $\mu_{q,i}$:

$$\mu_{q,i} = -\frac{1 + a\psi}{\psi q_i} - \frac{\log q_i}{\psi} + \left(\frac{1}{\psi} + \delta\right) - \sigma(\sigma_q)_{ii} + \frac{\|\sigma^K_{i\cdot}\|^2}{\eta_i} + r.$$

The quadratic risk premium term driving financial amplification is

$$\mathrm{RP}_i = \frac{\|\sigma^K_{i\cdot}\|^2}{\eta_i}.$$

Risk compensation rises nonlinearly as expert wealth share η_i shrinks, generating powerful balance-sheet amplification of fundamental shocks.

2.4 State Laws of Motion for Wealth Shares

2.4.1 Expert wealth share η_i

$$d\eta_i = \left[\left(\frac{1 + a\psi}{\psi q_i} - \frac{1}{\psi} - \rho - \chi\right)\eta_i + \left(\frac{1}{\eta_i} - 1\right)^2 \eta_i \|\sigma^K_{i\cdot}\|^2\right] dt + (1 - \eta_i)\sigma^K_{i\cdot} d\boldsymbol{W}.$$

2.4.2 Household wealth share ζ_i

Define country-i household capital return drift μ^K_i and world portfolio aggregates:

$$\mu^K_i = -\frac{1 + a\psi}{\psi q_i} + \frac{1}{\psi} + \frac{\|\sigma^K_{i\cdot}\|^2}{\eta_i} + r, \quad \mu_H = \sum_j \zeta_j \mu^K_j, \quad \sigma_H = \sum_j \zeta_j \sigma^K_{j\cdot}.$$

The wealth share diffusion is

$$\frac{d\zeta_i}{\zeta_i} = \left[\mu^K_i - \mu_H - \sigma_H \left(\sigma^K_{i\cdot} - \sigma_H\right)^\top\right] dt + \left(\sigma^K_{i\cdot} - \sigma_H\right) d\boldsymbol{W}.$$

2.5 Market Clearing Condition

Final goods market clearing imposes an exact cross-country restriction on capital prices, enforced directly via network output transformation (no soft loss penalty), see equation (19) in Huang 2022:

$$\sum_{i=1}^{3} \frac{\zeta_i}{q_i} = \frac{1 + \psi\rho}{1 + a\psi}.$$

2.6 Three-Country Block BSDE Iteration Algorithm

The critical identification principle: BSDE evaluation is well-conditioned only when all drift coefficients and state transition laws are fixed; simultaneous optimization over asset prices and the global interest rate destabilizes training. We implement a nonlinear block Gauss–Seidel alternating minimization scheme with two core steps per outer iteration:

2.6.1 Step A: Fixed $r(S)$ — BSDE Evaluation of Capital Prices $q_i, \sigma_{q,i\cdot}$

Given an initial guess for the interest rate function (baseline $r(S) = 0.03$), simulate forward state trajectories $S^{+,d}$ over independent Brownian shock batches. Regress future capital prices on current states and shock increments to recover BSDE intercept and slope terms:

$$q_i(S^{+,d}) = \alpha_i(S) + \beta_{i\cdot}(S)\Delta \boldsymbol{W}^d + \varepsilon_i^d.$$

$S^{+,d}$ is the set of states $(\eta_{1,t+\Delta}, \eta_{2,t+\Delta}, \eta_{3,t+\Delta}, \zeta_{1,t+\Delta}, \zeta_{1,t+\Delta})$, after the one step update from time t. The drift and volatility terms used to update state variables are guessed by network, risk free rate is frozen. BSDE consistency requires

$$\alpha_i = q_i + q_i\mu_{q,i}\Delta t, \qquad \beta_{i\cdot} = q_i\sigma_{q,i\cdot}.$$

The normalized training loss for the price-volatility network is

$$\mathcal{L}_q = \frac{\left\|\alpha - q - q\mu_q\Delta t\right\|^2}{\mathbb{E}[\mu_q^2]} + \frac{\left\|\beta/q - \sigma_q\right\|^2}{\frac{1}{2}\mathbb{E}\left[\|\sigma_q\|^2 + \|\beta/q\|^2\right]},$$

with detached normalization constants to prevent trivial scale-based loss reduction. Given a state S, we simulate numerous Brownian motion paths d, and the α and β estimated via OLS are relatively accurate.

2.6.2 Step B: Fixed (q, σ_q) — Update Global Risk-Free Rate $r(S)$

Freeze the trained price-volatility network. Train a separate neural network for $r(S)$ against the identical BSDE conditional regression loss, solving for a single common interest rate consistent with all three countries' Euler drift restrictions.

The full outer block iteration mapping is

$$r^n \mapsto \left(q^{n+1}, \sigma_q^{n+1}\right) \mapsto r^{n+1}.$$

Given the n-th iteration interest rate r^n, we run OLS and solved $\alpha_1^n, \alpha_2^n, \alpha_3^n$ for three asset prices. While the dynamics of three assets may not imply a exactly same global interest rate. We minimize the loss:

$$\min_{r^{n+1}} \sum_{i=1}^{3} (\alpha_i^n - q_i^n - q_i^n \mu^q(q^n; \sigma_q^n; r^{n+1}))^2$$

We aim to ensure that the global risk implied by the asset-price BSDE trained from $r(S)$ reverts back to $r(S)$, through which we obtain the fixed point.

2.7 Structural Parameter Comparative Results

Table 2: Key Structural Calibration

Parameter	Baseline Value
Productivity a	0.10
Depreciation δ	0.05
Fundamental volatility σ	0.023
Preference curvature ψ	5.0

2.7.1 Varying fundamental shock volatility $\sigma \in \{0.015, 0.023, 0.035\}$

This paper checks different fundamental shock volatility $\sigma \in \{0.015, 0.023, 0.035\}$. We solve the model under each volatility parameter and terminate alternating optimization training once the outer-round loss sequence plateaus (i.e., loss no longer exhibits meaningful downward decay).

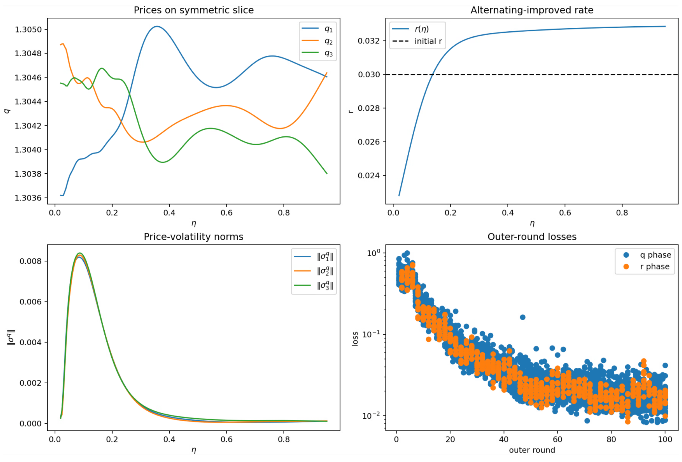
For all three calibration cases, we impose symmetric wealth weights $\zeta_1 = \zeta_2 = \zeta_3 = 1/3$, which implies identical steady-state wealth shares across the three country agents. We fix the expert weight parameters $\eta_1 = \eta_2 = \eta_3$ to examine equilibrium asset prices, price volatility norms, alternating-improved interest rates, and outer-round training loss curves under equalized expert wealth allocation, see Figure 2.

Taking the calibration with fundamental shock volatility $\sigma = 0.023$ as an illustrative example, we conduct a cross-BSDE consistency test to validate fixed-point convergence of the global interest rate. We extract implied interest rate sequences separately from each of the three asset price BSDE systems q_1, q_2, q_3, and compare them against the global rate $r(\eta)$ solved by our alternating optimization framework. As illustrated in Figure 3, the three implied rate curves tightly overlap with the network-solved global rate over the entire support of η. This uniform consistency across all three price equations strongly supports the claim that we have successfully located the unique fixed point of the global interest rate functional.

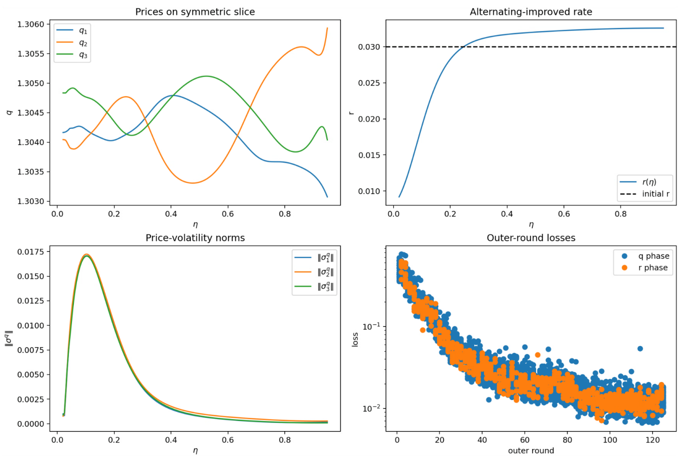
2.7.2 Economic Interpretation of the Solution

We maintain symmetric wealth weights $\zeta_1 = \zeta_2 = \zeta_3 = 1/3$ and identical expert allocation parameters $\eta_1 = \eta_2 = \eta_3$ as our baseline calibration. Even though households are precluded from holding capital entirely, our model recovers core qualitative patterns consistent with the mechanism laid out in Brunnermeier and Sannikov 2014.

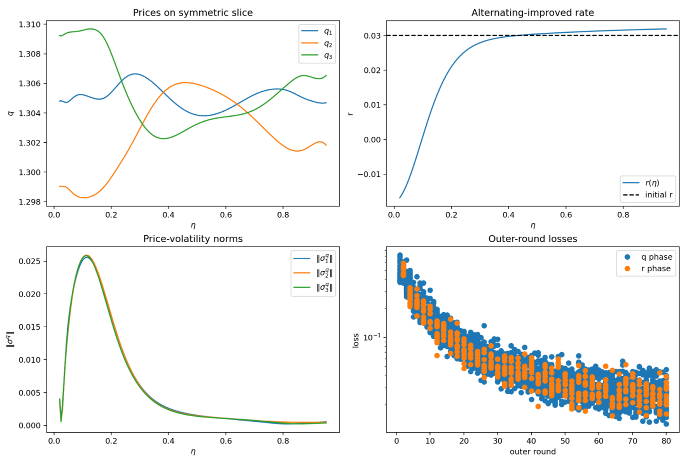
When η takes large values, experts operate with low financial leverage; aggregate fundamental shocks exert muted price volatility responses due to their substantial wealth buffer. Around $\eta \approx 0.2$, experts' leverage peaks sharply. Adverse fundamental shocks rapidly erode expert net wealth, triggering a dramatic surge in asset price volatilities captured by the volatility loadings $\sigma^q_{ij}(\eta)$. For very small η, savers flood the economy with excess supply of risk-free claims, pushing the equilibrium risk-free rate into negative territory. Although experts demand elevated risk compensation, a portion of this risk premium is offset by the stabilizing buffer from depressed interest rates, which restrains extreme asset volatility.



(a) $\sigma = 0.015$



(b) $\sigma = 0.023$



(c) $\sigma = 0.035$

Figure 2: Numerical equilibrium results under symmetric wealth weights $\zeta_1 = \zeta_2 = \zeta_3 = 1/3$ and identical expert allocation parameters $\eta_1 = \eta_2 = \eta_3$.

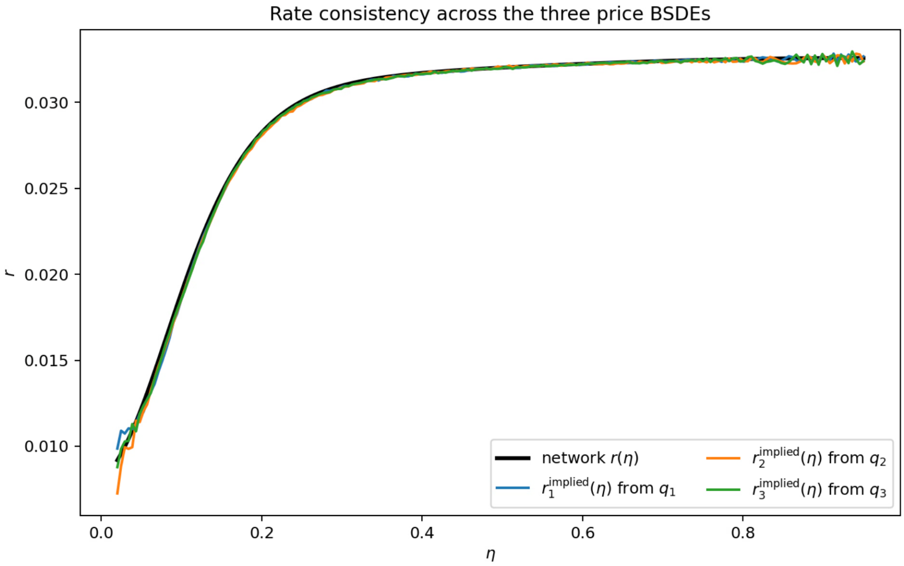


Figure 3: Consistency verification of equilibrium interest rates under $\sigma = 0.023$. The black solid line denotes the global rate function $r(\eta)$ solved directly via our alternating neural network algorithm. The blue, orange and green curves represent the implied interest rates $r_1^{\text{implied}}(\eta)$, $r_2^{\text{implied}}(\eta)$, $r_3^{\text{implied}}(\eta)$ recovered separately from the three asset price BSDEs q_1, q_2, q_3. The near-perfect overlap of all four curves across the full domain of η provides strong numerical evidence that we have converged to a consistent fixed point of the global interest rate functional.

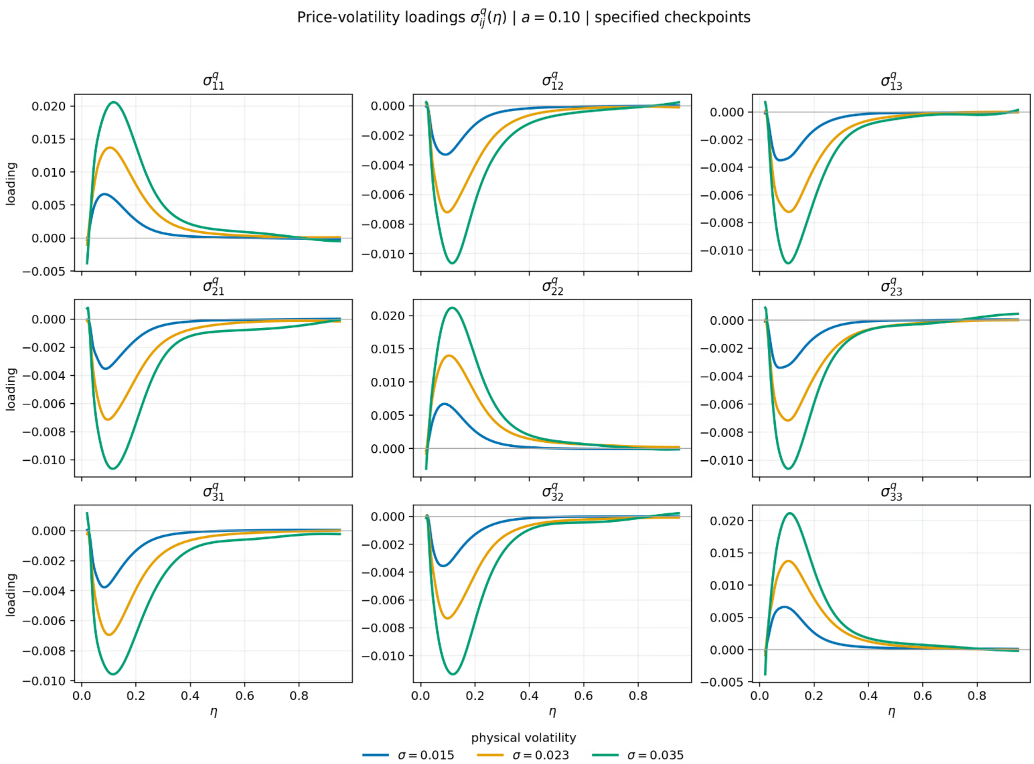


Figure 4: Matrix of price volatility loadings $\sigma^q_{ij}(\eta)$ under $a = 0.10$.

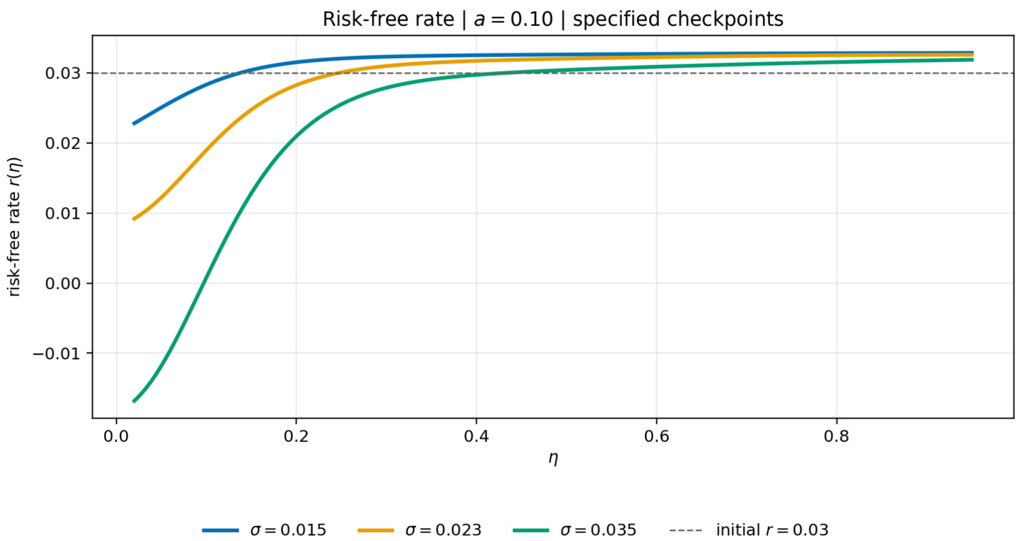


Figure 5: Equilibrium risk-free rate $r(\eta)$ under $a = 0.10$.

2.7.3 Varying production efficiency $a \in \{0.04, 0.07, 0.10\}$

Fix fundamental volatility $\sigma = 0.023$. Let an equilibrium triple $(q_i^0, \sigma_{q,i\cdot}^0, r^0)$ correspond to efficiency a_0, and define scaling factor

$$\lambda = \frac{1 + a_1\psi}{1 + a_0\psi}.$$

Construct transformed equilibrium objects for alternative a_1:

$$q_i^1 = \lambda q_i^0, \quad \sigma_{q,i\cdot}^1 = \sigma_{q,i\cdot}^0, \quad r^1 = r^0 + \frac{\log \lambda}{\psi}.$$

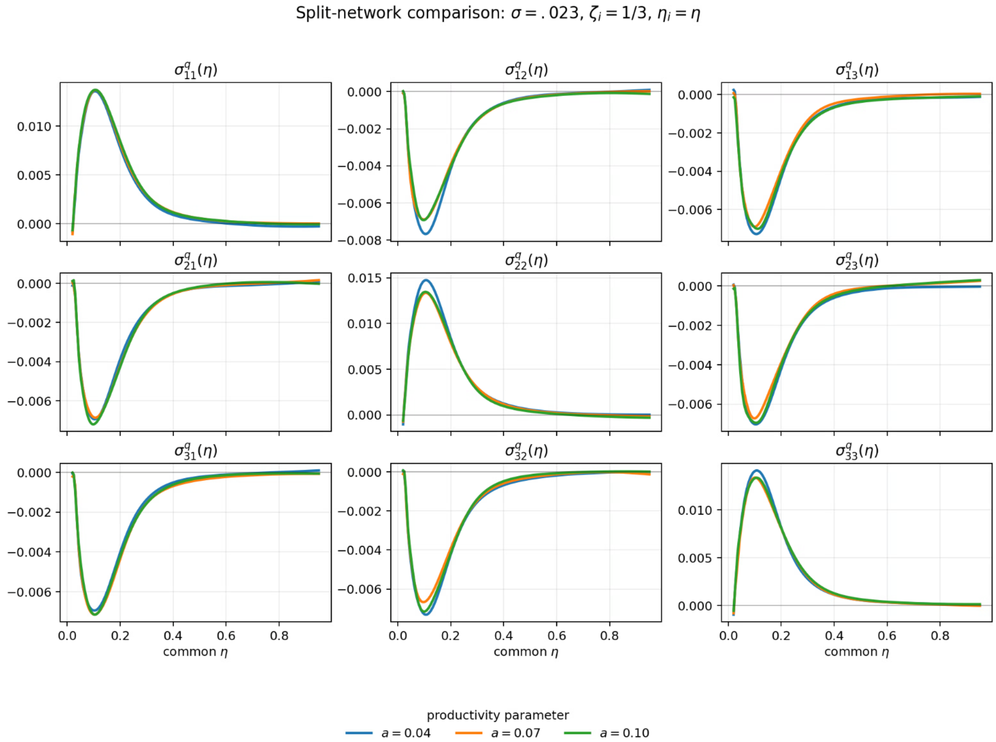
Investment and capital growth satisfy

$$a_1 - \iota_i^1 = \lambda(a_0 - \iota_i^0) \implies \frac{a_1 - \iota_i^1}{q_i^1} = \frac{a_0 - \iota_i^0}{q_i^0}.$$

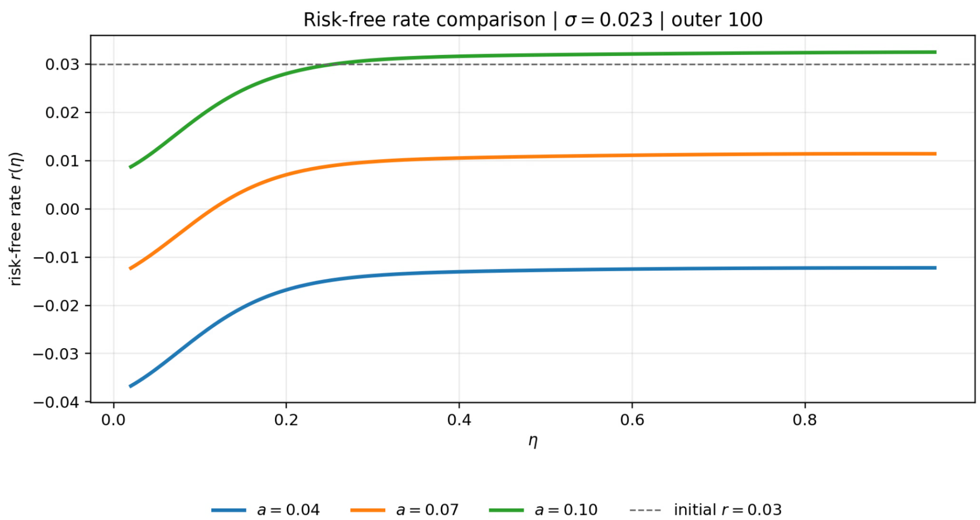
All state transition dynamics remain unchanged, and the price diffusion process satisfies

$$\frac{dq_i^1}{q_i^1} = \frac{dq_i^0}{q_i^0} \implies \boxed{\sigma_q^{a_1}(S) = \sigma_q^{a_0}(S)}.$$

Key implication: Shifts in production efficiency a scale the level of capital prices and the global risk-free rate, but leave relative price volatility σ_q unchanged. The absolute volatility exposure $q_i\sigma_{q,i\cdot}$ scales proportionally with q_i.



(a) Price volatility loadings under varying productivity parameter a



(b) Equilibrium risk-free rate for alternative productivity values a

Figure 6: Robustness test with alternative values of the productivity parameter a. We fix $\sigma = 0.023$, symmetric wealth weights $\zeta_i = 1/3$ and equal expert allocation parameters $\eta_i = \eta$. Panel (a) shows that changes in a generate minor deviations in the price volatility loadings $\sigma^q_{ij}(\eta)$. Curves corresponding to $a = 0.04$, $a = 0.07$ and $a = 0.10$ closely coincide, verifying the numerical robustness of our solution algorithm. Panel (b) illustrates the equilibrium risk-free rate $r(\eta)$: a higher a yields a higher risk-free rate, and interest rate curves exhibit near-parallel vertical shifts. Raising a lifts the entire schedule of $r(\eta)$ while barely changing the curve shape over η. The dashed horizontal line marks the exogenous initial reference rate $r = 0.03$.

References

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